

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 2

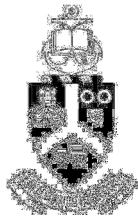
Kopiereg voorbehou

Module EBN122
Elektrisiteit en Elektronika
7 Oktober 2013

Semester Test 2

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Module EBN122
Electricity and Electronics
7 October 2013



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute <i>90 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		14	

BELANGRIK- IMPORTANT

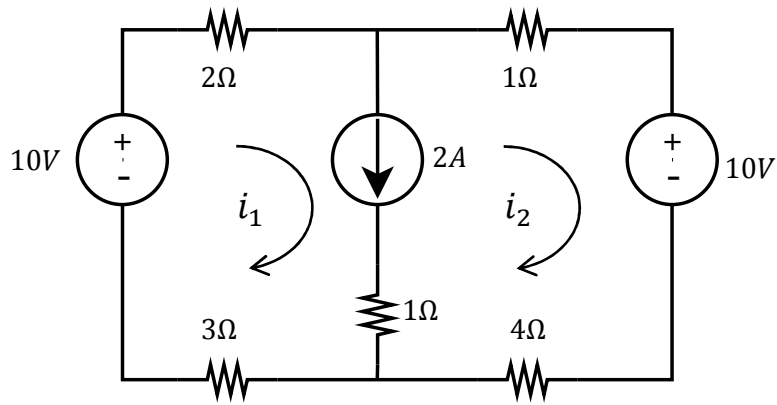
1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 JD le Roux

VRAAG/QUESTION 1 [10]

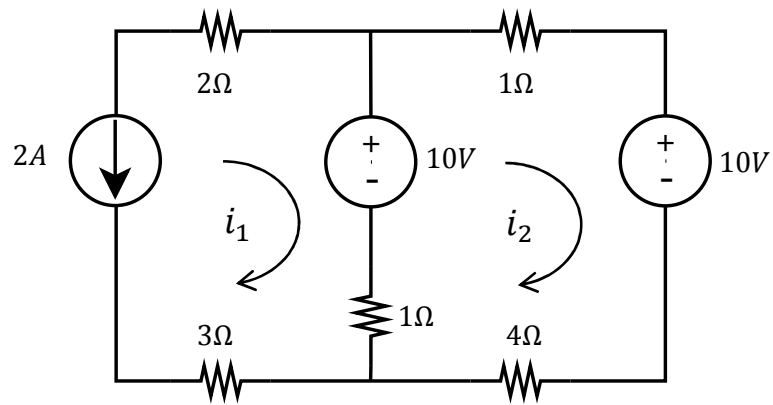
Vraag 1.1 [2]
Bepaal i_1 en i_2 .

Question 1.1 [2]
Determine i_1 and i_2 .



Vraag 1.2 [2]
Bepaal i_1 en i_2 .

Question 1.2 [2]
Determine i_1 and i_2 .

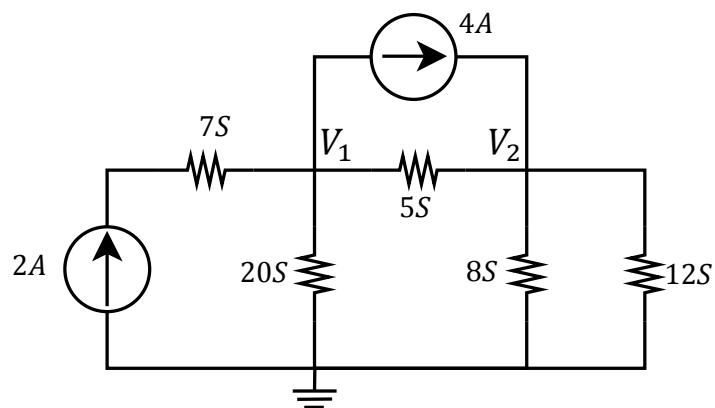


Vraag 1.3 [2]

Stel 'n matriksvergelyking d.m.v. inspeksie op in terme van die nodespannings.

Question 1.3 [2]

Use inspection to set up a matrix equation in terms of the nodal voltages.



Vraag 1.4 [4]

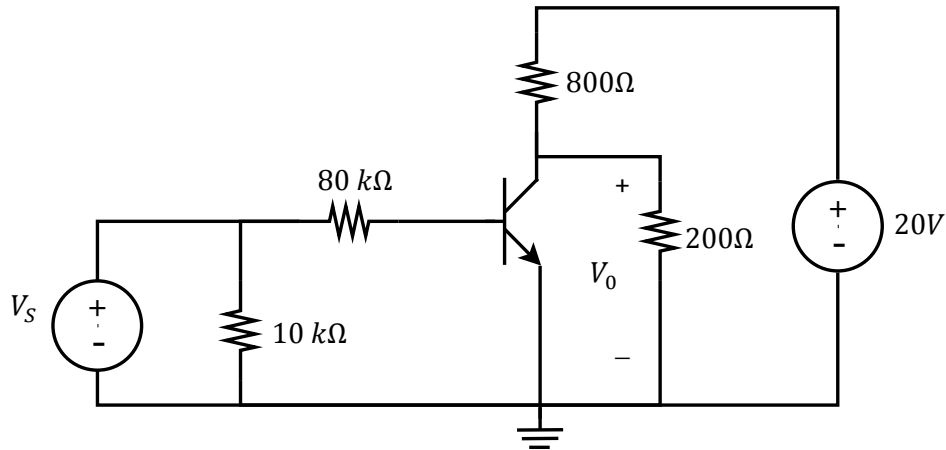
Gegee: $\beta = 100$ en $V_{BE} = 0.7\text{ V}$.

- a) Indien $V_S = 10.7$, bepaal I_B .
- b) Indien $I_B = 0.1\text{ mA}$, bepaal V_0 .

Question 1.4 [4]

Given: $\beta = 100$ and $V_{BE} = 0.7\text{ V}$.

- a) If $V_S = 10.7$, determine I_B .
- b) If $I_B = 0.1\text{ mA}$, determine V_0 .



VRAAG/QUESTION 2 [20]

Vraag 2.1 [4]

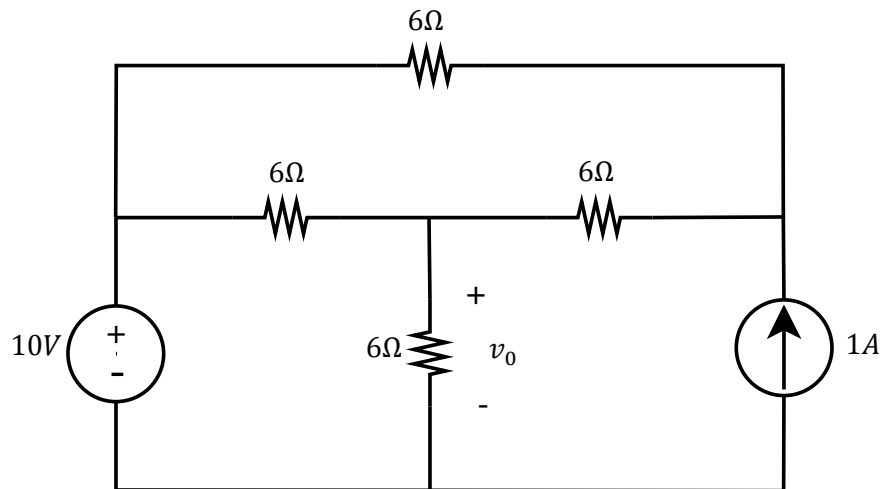
Gebruik superposisie:

- a) Bereken v_0 slegs agv die 10V spanningsbron.
- b) Bereken v_0 slegs agv die 1A stroombron.

Question 2.1 [4]

Use superposition:

- a) Calculate v_0 due to the 10V voltage source.
- b) Calculate v_0 due to the 1A current source.

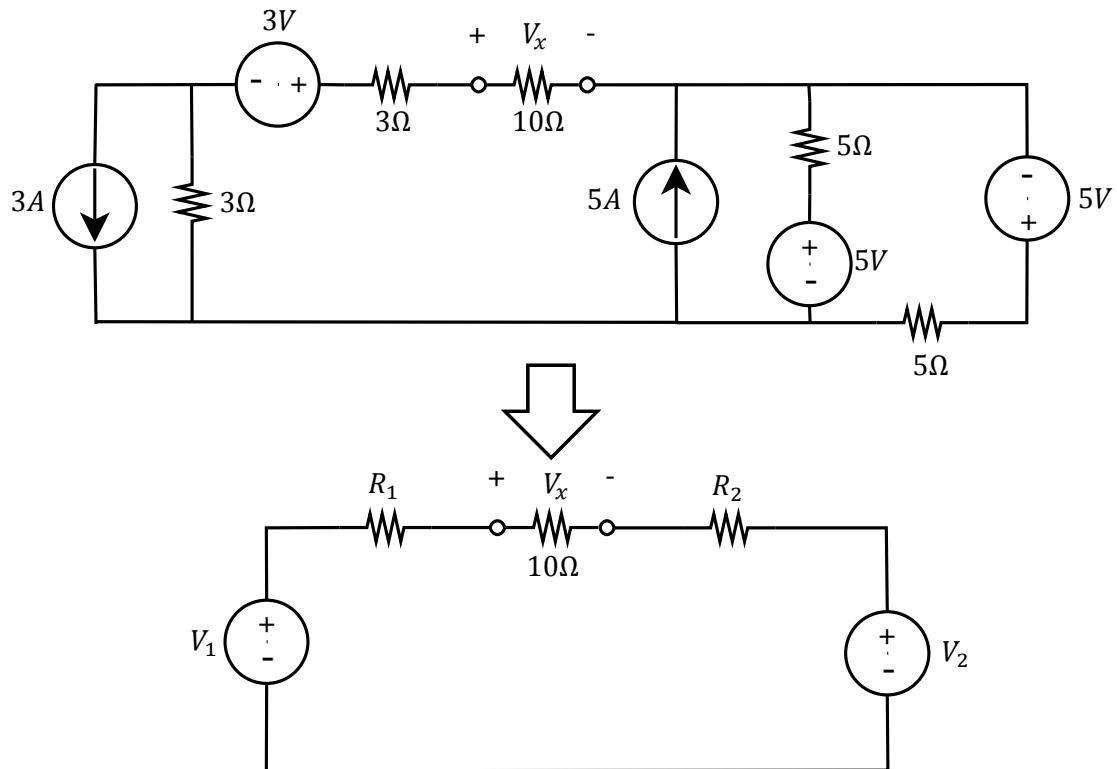


Vraag 2.2 [4]

Gebruik brontransformaties om die stroombaan te vereenvoudig. Bepaal V_1 , R_1 , V_2 en R_2 .

Question 2.2 [4]

Use source transformations to simplify the circuit below. Determine V_1 , R_1 , V_2 and R_2 .

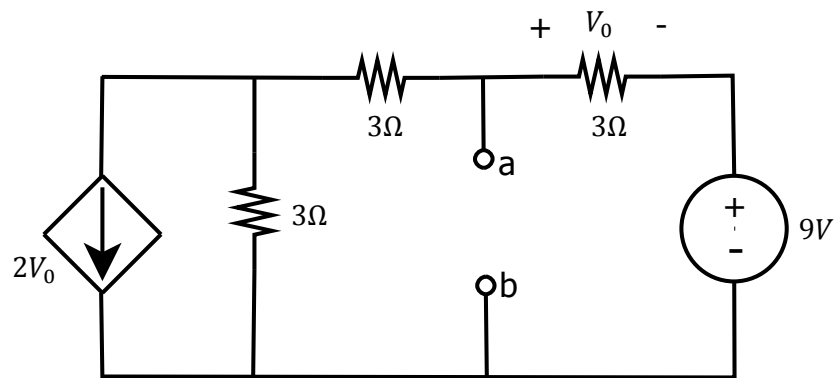


Vraag 2.3 [4]

Bepaal die Thevenin ekwivalente stroombaan tov terminale a-b.

Question 2.3 [4]

Determine the Thevenin equivalent circuit wrt terminals a-b.

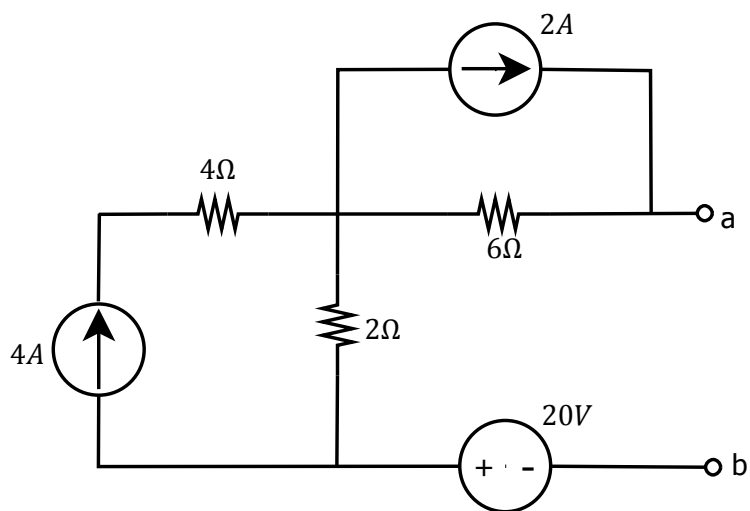


Vraag 2.4 [4]

Bepaal die Norton ekwivalente stroombaan tov terminale a-b.

Question 2.4 [4]

Determine the Norton equivalent circuit wrt terminals a-b.



Vraag 2.5 [4]

- a) 'n Battery het 'n kortsluitstroom van 20A en 'n opebaan spanning van 12V. As die battery oor 'n 2Ω weerstand gekoppel word, bereken die drywing wat in die 2Ω weerstand verkwis sal word.
- b) Beskou die stroombaan van Vraag 2.1: die lasweerstand is die 6Ω weerstand waaroor die uitset v_0 gedefinieer is. Met watter ander waarde weerstand sal jy die lasweerstand vervang as jy maksimum drywing wil verkwis in die las.

Question 2.5 [4]

- a) A battery has a short circuit current of 20A and an open-circuit voltage of 12V. If the battery is connected across a 2Ω resistor, calculate the power dissipated by the 2Ω resistor.
- b) Consider the circuit of Question 2.1: the load resistor is the 6Ω resistor over which the output v_0 is defined. With which other value resistor will you replace the load resistor if you want maximum power dissipation in the load.

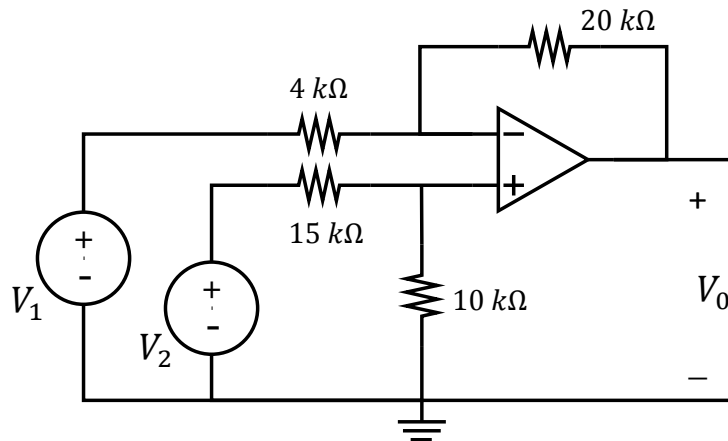
VRAAG/QUESTION 3 [15]

Vraag 3.1 [4]

- a) Indien $V_1 = 0$ V en $V_2 = 1$ V, bepaal V_0 .
b) Indien $V_1 = 1$ V en $V_2 = 0$ V, bepaal V_0 .

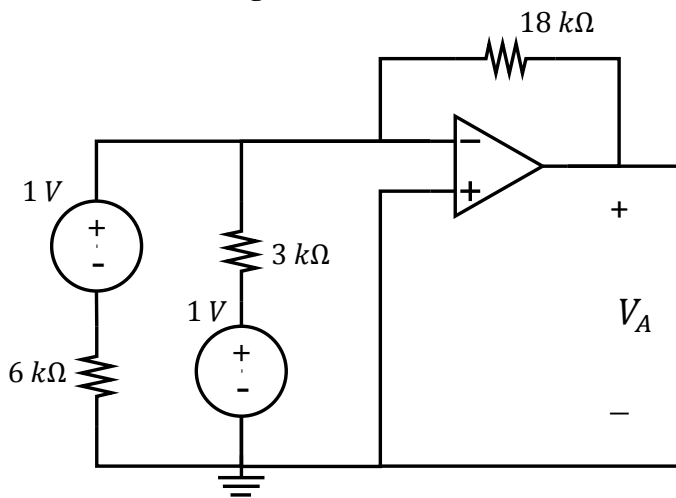
Question 3.1 [4]

- a) If $V_1 = 0$ V and $V_2 = 1$ V, determine V_0 .
b) If $V_1 = 1$ V and $V_2 = 0$ V, determine V_0 .

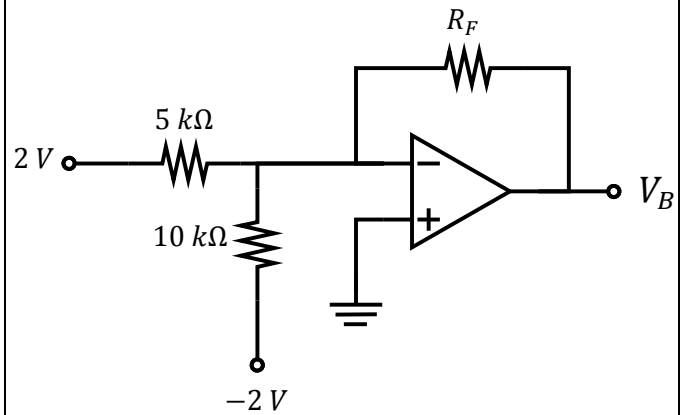


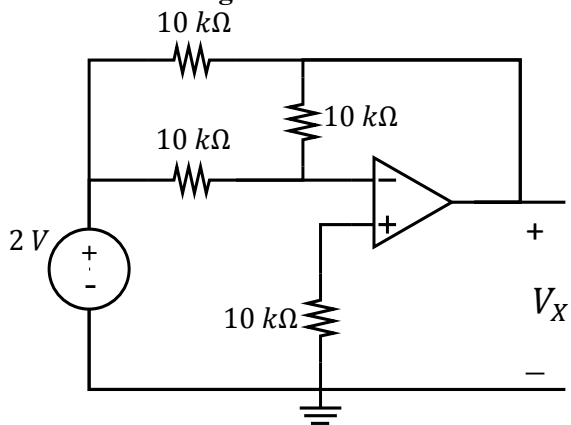
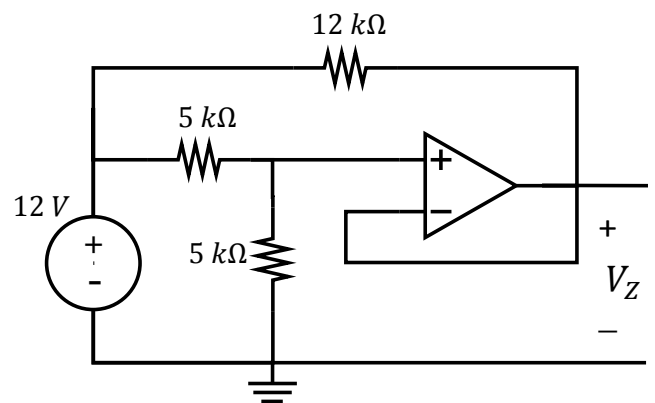
Vraag 3.2 [4]

- a) Kringbaan A: Bepaal V_A .
b) Kringbaan B: Bepaal R_F indien $V_B = -4\text{ V}$.

Kringbaan A / Circuit A**Question 3.2 [4]**

- a) Circuit A: Determine V_A .
b) Circuit B: Determine R_F if $V_B = -4\text{ V}$.

Kringbaan B / Circuit B

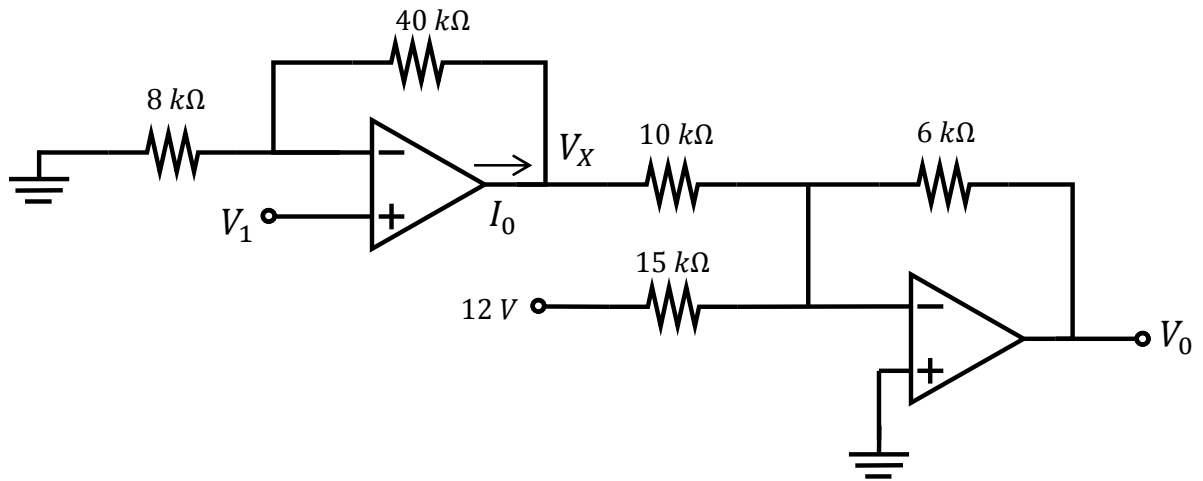
Vraag 3.3 [4]a) Kringbaan A: Bepaal V_X .b) Kringbaan B: Bepaal V_Z .**Kringbaan A / Circuit A****Question 3.3 [4]**a) Circuit A: Determine V_X .b) Circuit B: Determine V_Z .**Kringbaan B / Circuit B**

Vraag 3.4 [3]

Indien $V_X = 15\text{ V}$, bepaal V_I , V_O , en I_O .

Question 3.4 [3]

If $V_X = 15\text{ V}$, determine V_I , V_O , and I_O .



EBN122 SEMESTER TOETS/TEST 2 (07-10-2013)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:		Studentenommer: Student number:						
Voorletters / Initials:		Taal / Language:	Afr		Eng			

Q1		Q2		Q3		Tot:	/45
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VRAAG 1 – QUESTION 1 (10)		
1.1	$i_1 =$	$i_2 =$
1.2	$i_1 =$	$i_2 =$
1.3	$\begin{bmatrix} \text{---} & \text{---} \\ \text{---} & \text{---} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} \text{---} \\ \text{---} \end{bmatrix}$	
1.4	$a) I_B =$	$b) V_o =$

VRAAG 2 – QUESTION 2 (20)			
2.1	$v_{o(10V)} =$		$v_{o(1A)} =$
2.2	$V_I =$	$R_I =$	$V_2 =$ $R_2 =$
2.3	$V_{Th} =$		$R_{Th} =$
2.4	$I_N =$		$R_N =$
2.5	$P =$		$R =$

VRAAG 3 – QUESTION 3 (15)			
3.1	$a) V_o =$		$b) V_o =$
3.2	$a) V_A =$		$b) R_F =$
3.3	$a) V_X =$		$b) V_Z =$
3.4	$V_I =$	$V_o =$	$I_o =$